IOT • AUTOMATED ATTENDANCE

RFID Attendance System

Using NodeMCU (ESP8266) and Google Sheets

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INTRODUCTION

Why an RFID Attendance System?

1

Manual attendance is slow

Roll calls and paper registers waste class/work time and are easy to falsify.

2

RFID enables instant identification

Each person carries a unique RFID tag that is read in a fraction of a second.

3

Cloud logging removes paperwork

Scan data is pushed straight to a Google Sheet — accessible from anywhere, on any device.

4

Low-cost, beginner-friendly hardware

Built entirely on affordable, widely available IoT components.

SYSTEM OVERVIEW

How the System Works

1. **RFID Tag** — User taps card / keyfob
2. **RC522 Reader** — Reads the unique tag UID
3. **NodeMCU ESP8266** — Verifies UID & timestamps
4. **Wi-Fi Network** — Sends data over HTTP(S)
5. **Google Sheets** — Logs name, date & time

The reader continuously polls for tags; on a valid scan the NodeMCU firmware matches the UID to a stored card-holder name, builds a timestamp, and calls a Google Apps Script Web App URL that appends a new row to the spreadsheet.

HARDWARE

Components Used

1

NodeMCU ESP8266

Wi-Fi enabled microcontroller — the brain of the system, handles logic & connectivity

2

RFID-RC522 Module

13.56 MHz reader/writer that scans MIFARE RFID tags and key fobs via SPI

3

RFID Tags / Cards

Passive tags, each holding a unique, factory-set UID

4

Buzzer

Gives instant audio feedback on a successful / failed scan

5

Breadboard & Jumper Wires

For prototyping the circuit without soldering

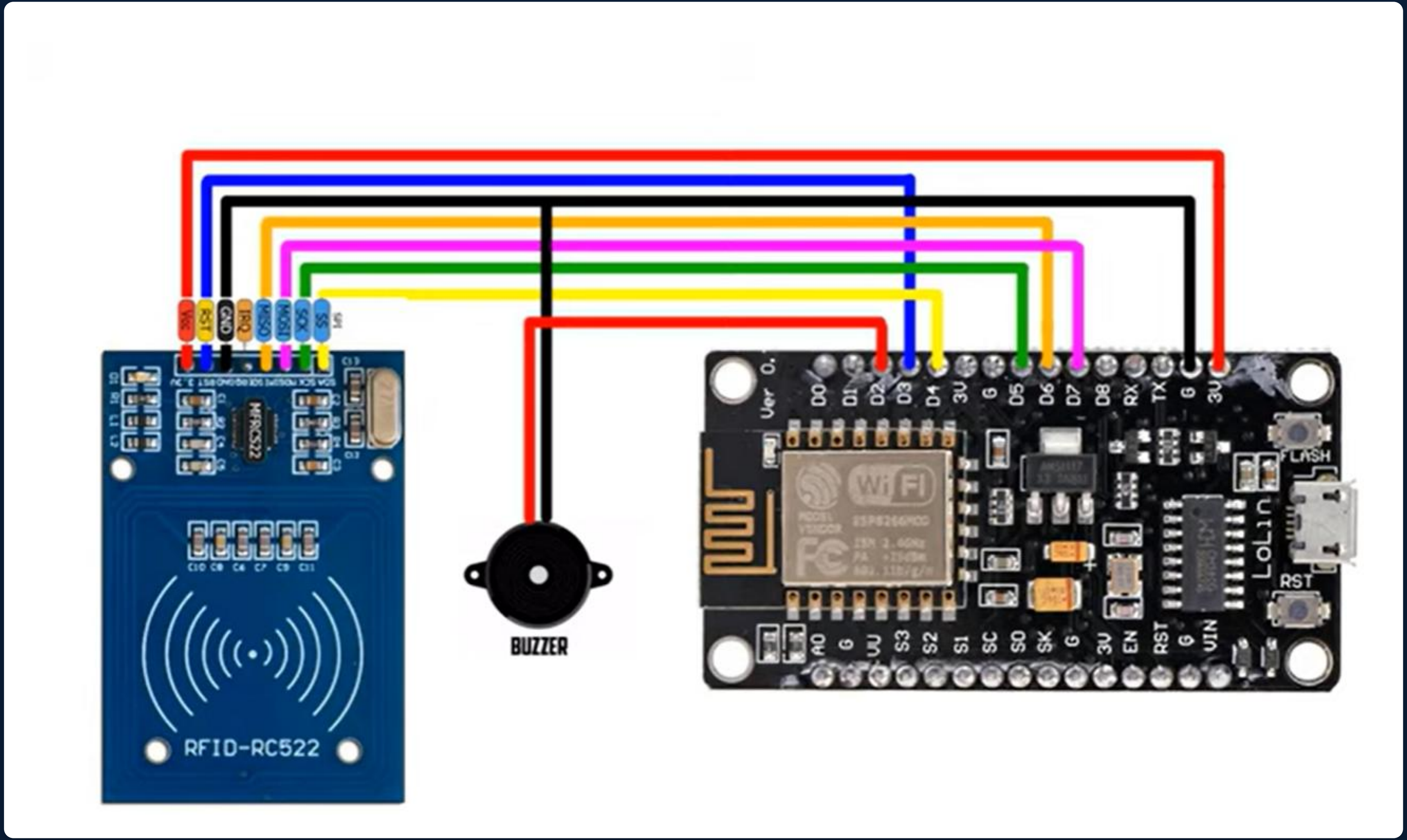
6

USB Power Supply

Powers the NodeMCU during operation

WIRING

Circuit Diagram

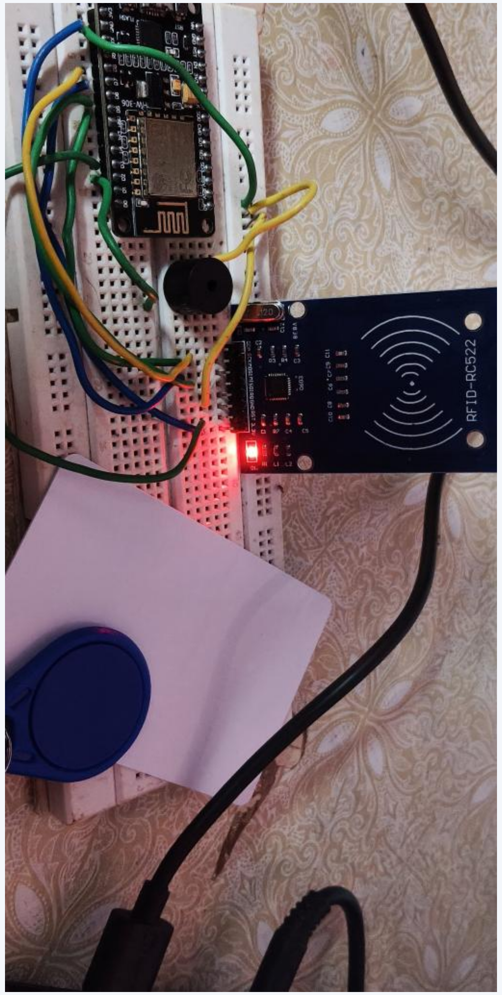


RC522 → NodeMCU

- **RST** GPIO 2 (D4)
- **SDA / SS** GPIO 15 (D8)
- **MOSI** GPIO 13 (D7)
- **MISO** GPIO 12 (D6)
- **SCK** GPIO 14 (D5)
- **3V3 / GND** Power & Ground

PROTOTYPE

Hardware Setup



Breadboard prototyping

All modules are wired on a breadboard for quick iteration before a permanent build.

Live status LED

The RC522's onboard LED lights up red on every successful tag read.

Tap-to-scan

A key-fob and card are shown being scanned within range of the RC522 antenna.

Compact footprint

NodeMCU, RC522, and buzzer fit comfortably on a single small breadboard.

SOFTWARE STACK

From Firmware to Spreadsheet

01

Arduino IDE

Firmware for the NodeMCU is written in C++ using the MFRC522 and ESP8266WiFi libraries.

02

Wi-Fi + HTTPClient

The board connects to a Wi-Fi network and sends an HTTPS GET/POST request per scan.

03

Google Apps Script

A deployed Web App receives the request and parses the UID / name / timestamp.

04

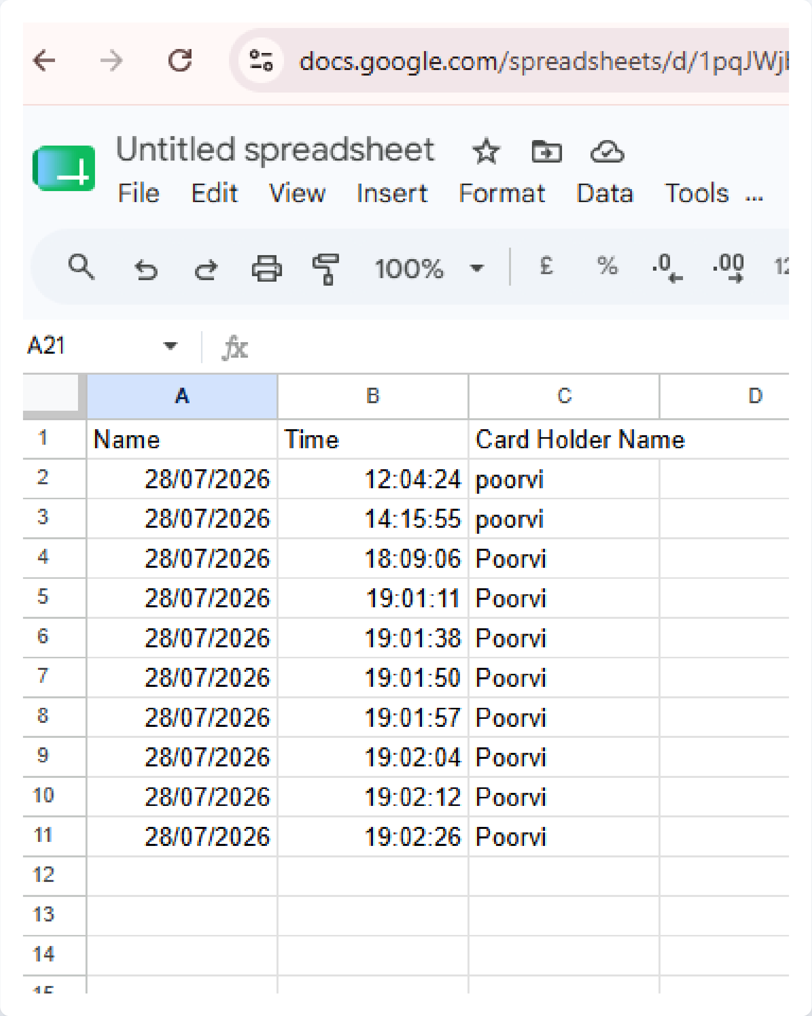
Google Sheets API

Apps Script appends a new row — no database server required.

Each row written to the sheet captures three fields — Name, Time, and Card Holder Name — giving an auditable, timestamped attendance log that the whole team can view live in Google Sheets.

RESULT

Live Attendance Log in Google Sheets



	A	B	C	D
1	Name	Time	Card Holder Name	
2	28/07/2026	12:04:24	poorvi	
3	28/07/2026	14:15:55	poorvi	
4	28/07/2026	18:09:06	Poorvi	
5	28/07/2026	19:01:11	Poorvi	
6	28/07/2026	19:01:38	Poorvi	
7	28/07/2026	19:01:50	Poorvi	
8	28/07/2026	19:01:57	Poorvi	
9	28/07/2026	19:02:04	Poorvi	
10	28/07/2026	19:02:12	Poorvi	
11	28/07/2026	19:02:26	Poorvi	
12				
13				
14				

What gets logged

Date

Automatically stamped by the script on every scan

Time

Recorded to the second, so repeat scans are traceable

Card Holder Name

Resolved from the tag's UID via a lookup table in firmware

Sample rows above show multiple scans for the same card holder — each tap appends a fresh, independent record.

HIGHLIGHTS

Key Features & Advantages

Real-time logging

Attendance appears in the sheet within seconds of a tap.

Contactless & fast

No manual entry — a single tap identifies the person.

Cloud accessible

View or export attendance from any device, anywhere.

No database needed

Google Sheets acts as a free, zero-maintenance backend.

Low cost

Built from inexpensive, off-the-shelf IoT components.

Easy to scale

New tags and users can be added in firmware in minutes.

WHERE IT FITS

Applications & Future Scope

APPLICATIONS

1. Schools & colleges — student attendance tracking
2. Offices — employee clock-in / clock-out logs
3. Labs & libraries — controlled equipment / room access
4. Events — quick check-in for registered attendees

FUTURE SCOPE

1. Add a small display (OLED/LCD) for instant scan feedback
2. Send email / SMS / app notifications on each scan
3. Add facial recognition as a second verification factor
4. Build an admin dashboard for analytics and reports

CONCLUSION

A Practical, Low-Cost Path to Smart Attendance

This project shows how a simple RFID reader, a Wi-Fi microcontroller, and a free Google Sheet can replace manual attendance registers with an instant, tamper-resistant, cloud-logged system — built for under the cost of a textbook.

Thank You